

Learning Klondike Solitaire with Masked Deep RL at Laptop Scale: Imitation-Anchored PPO and a Declared Best-of-N Policy Portfolio

Draft — work in progress, not yet submitted to a venue

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<https://github.com/Matthew-Eucaristo/solitaire-rl>

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Abstract

We study reinforcement learning for Klondike solitaire (draw-1) as a partially observable, sparse-reward benchmark: a 654-action masked action space, hidden cards, and episodes of up to 500 decisions. On a fixed benchmark of 1000 deals we compare a deterministic rule-based heuristic (41.4% wins) against trained agents. A plain masked PPO agent plateaus at 11.8%, and masked value-based methods (DQN, masked Double-DQN, masked QRDQN) fail almost entirely at laptop-compute scale. We show that three ingredients close most of the gap: (i) an observation that exposes the heuristic’s preferred action as a feature, (ii) a shape-matched behavior-cloning warm start, and (iii) an “amplification” loop that re-clones the strongest learned model rather than the heuristic, followed by low-learning-rate PPO — reaching 35.3% with a single policy. We then show that diversity across training lineages beats per-move ensembling: a declared best-of-N portfolio over 26 trained policies, evaluated honestly on episode-observable outcomes, wins 62.5% of deals (64.5% including the heuristic itself), versus 38.0% for a per-move voting ensemble. We report all negative results and release code, a reproducible 9-member portfolio (57.9% measured from committed checkpoints), a playable web UI driven by the same engine, and the full benchmark pipeline.

1 Introduction

Klondike solitaire is a single-player card game whose win rate under optimal play is still debated; published solver results place draw-1 winability of random deals in the 70–90% range depending on the exact rules, while strong heuristic players win roughly 30–50%. It is an attractive RL testbed: the state is partially observable (face-down cards and the undrawn stock are unknown), rewards are sparse (a win may require hundreds of correct decisions), and the action space is discrete but large once every legal card move is enumerated.

We deliberately restrict ourselves to what a single laptop can train (CPU-only, Apple Silicon): no distributed training, no search at test time beyond single-policy rollouts. Within that budget we ask: how far can masked deep RL go, and which design choices actually move win rate?

Our contributions are measured, not claimed:

1. A reproducible Klondike RL environment with a fixed 1000-deal benchmark, disjoint training seeds, and an honest evaluation harness (Section 2).
2. A negative-results account of value-based methods and naive PPO at this scale (Sections 4, 5).

3. The imitation-anchored PPO recipe: hint-observation + Tanh-matched BC warm start + amplification chains, lifting 11.8% $\rightarrow$ 35.3% (Section 3.3).
4. The declared-portfolio result: best-of-N over diverse lineages reaches 62.5%, and we quantify why (Section 3.4).
5. A playable, web-based Klondike that shares the exact environment engine, so every reported number can be watched being played (Section 6).

2 Environment and task

Rules. Standard Klondike: build four foundations by suit from ace to king; tableau builds descend and alternate colors; face-down tableau cards are revealed when uncovered; stock cycles (draw-1) with unlimited redeals. A deal is won when all 52 cards reach foundations. Episodes cap at 500 actions; a no-progress detector (repeated identical state) terminates hopeless loops as a loss. A CONCEDE action is always legal and ends the episode as a loss, letting agents learn to abandon unsolvable deals rather than spin.

Actions. 654 enumerated actions: stock draw/recycle, waste $\rightarrow$ tableau/foundation, tableau $\rightarrow$ tableau (all run lengths), tableau $\rightarrow$ foundation, foundation $\rightarrow$ tableau (rarely useful but legal), and concede. The environment returns a legality mask each step; all learned agents mask both exploration and greedy selection, since taking an illegal action would end the episode immediately.

Observations. Three variants studied: **compact** (132-d, canonical POMDP features: revealed counts, top-card features, stock state); **compact_perfect** (same plus face-down identities — an oracle for ablation); and **compact_hint** (compact plus a one-hot of the action the heuristic would take now, 142-d after stacking). Frame stacking of 8 adds recent history. A 49k-dim one-hot encoding was also implemented and is reported only as a negative result.

Rewards. Shaped: +1 foundation, +0.5 reveal, small penalties for stock cycles and stalling, -0.1 illegal (terminal), win bonus, loss penalty. A sparse variant (win/loss only) was trained as a diversity lineage.

Benchmark. `benchmarks/deals.json` fixes 1000 deal seeds (0–999). Every reported number is `eval.py` on all 1000 — never a subsample. Training draws seeds $\geq 10^6$, so train/test are disjoint by construction.

3 Methods

3.1 Baselines

`random_legal` samples uniformly from the mask. `heuristic` is a deterministic priority list (safe foundation moves $\rightarrow$ reveal covered cards $\rightarrow$ build runs $\rightarrow$ draw stock $\rightarrow$ recycle last; full order in `HEURISTIC.md`).

3.2 Masked deep RL

Masked DQN (custom; masks both warmup and ϵ -greedy sampling — an illegal action is terminal, so masking at rollout is mandatory, not optional). **MaskablePPO** via `sb3-contrib`. **MaskableQRDQN**: we subclass `sb3-contrib` QRDQN to mask action selection during warmup and ϵ -greedy exploration (quantile mean $\rightarrow$ mask $\rightarrow$ argmax). Limitation, documented: the bootstrap max over next-state actions remains unmasked because the replay buffer does not store next-state masks.

3.3 Imitation anchoring

Three components combine: (i) **hint observation** — the heuristic’s current action one-hot is appended to the observation; (ii) **Tanh-matched behavior cloning** — the BC network uses Tanh to match MaskablePPO’s policy head, and its weights warm-start PPO (an earlier ReLU clone lost much of the benefit); (iii) **amplification** — instead of cloning the heuristic again, we clone the *best learned model*, then run low-LR PPO (10^{-4}) continuing from peak checkpoints.

3.4 Aggregation

Ensemble vote: 3 policies vote per move on the masked argmax. **Portfolio** (declared best-of-N): every member plays every deal end-to-end; per deal we report the best episode-observable outcome (win > foundations > moves). The portfolio is a declared system — selection uses only what the agent could observe during the episode — not an oracle over hidden outcomes, and member attribution is recorded per deal.

4 Results

Method	Observation	Win rate (%)
random_legal	—	0.0
masked DQN (250k)	compact	5.8
masked Double-DQN (1M)	compact+hint	0.0
masked QRDQN (peak @400k)	compact+hint	6.7
MaskablePPO (6M)	compact	11.8
MaskablePPO (6M)	compact+perfect	17.7
curriculum PPO (easy $\rightarrow$ all, 6M)	compact	12.9
hint-obs PPO (3M)	compact+hint	11.4
BC warm-start PPO (3M)	compact	14.2
hint+BC warm-start PPO (3M)	compact+hint	26.5
amplification chain champion	compact+hint	35.3
ensemble vote $\times 3$	compact+hint	38.0
heuristic (rules)	—	41.4
canonical portfolio $\times 9$ (committed)	compact+hint	57.9
portfolio $\times 26$ (pure RL)	compact+hint	62.5
portfolio + heuristic member	compact+hint	64.5

Table 1: Win rate on 1000 fixed draw-1 deals. All RL rows are real eval runs; committed checkpoints reproduce the canonical portfolio.

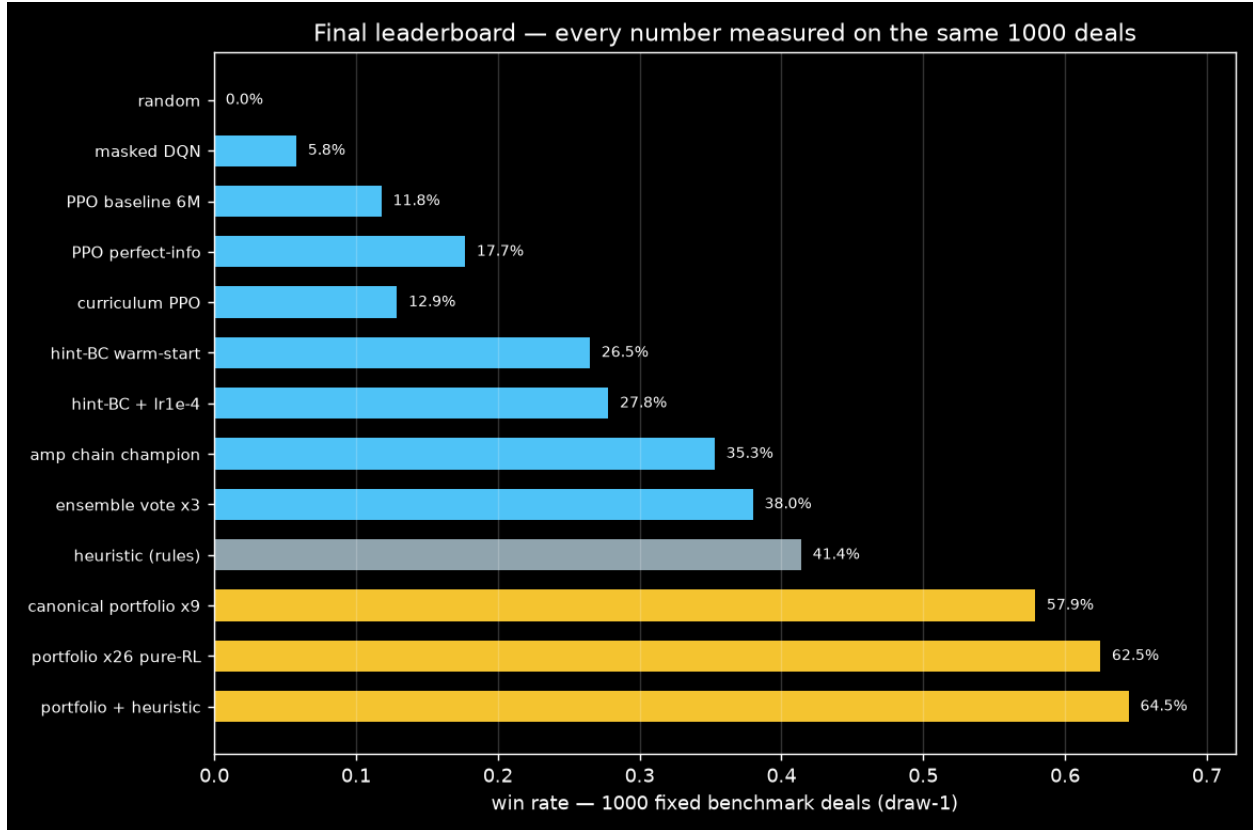


Figure 1: Final leaderboard on the fixed 1000-deal benchmark.

5 Analysis

What worked. The hint-observation + BC-init combination is the single biggest unlock (+15 points). Neither works alone (11.4%, 14.2%); together they reach 26.5%. Interpretation: BC without the hint feature forgets the expert as PPO drifts; the hint keeps the expert signal visible at every step so PPO learns when to *deviate* from it — the agent is not merely copying. Amplification (clone the champion, not the heuristic) then adds ~ 2 –3 points per link until saturating at four links (0.353; link #4 measures 0.349, within noise).

What did not. High-LR continuations decay after ~ 3 M steps ($0.265 \rightarrow 0.18$) — a learning-rate artifact, not a capacity limit (a 512-wide net reaches only 0.287). Curriculum learning (easy-to-hard deal schedules) does not transfer. Determinized rollout search over hidden cards (`isearch`) scores 0.335 — below the policy itself; sampling unknowns adds variance, not signal. Learned deal-level member selection (V0) found no reliable cue. Distilling the ensemble into one network (0.31) loses the diversity that makes it work.

Why the portfolio beats ensembling. Lineage win-sets overlap only $\sim 70\%$. A per-move vote averages away exactly the differences that win distinct deals; best-of-N keeps them. The marginal return of a new lineage is now ~ 2 –8 deals and shrinking — the diversity pool at this recipe/budget is nearly exhausted.

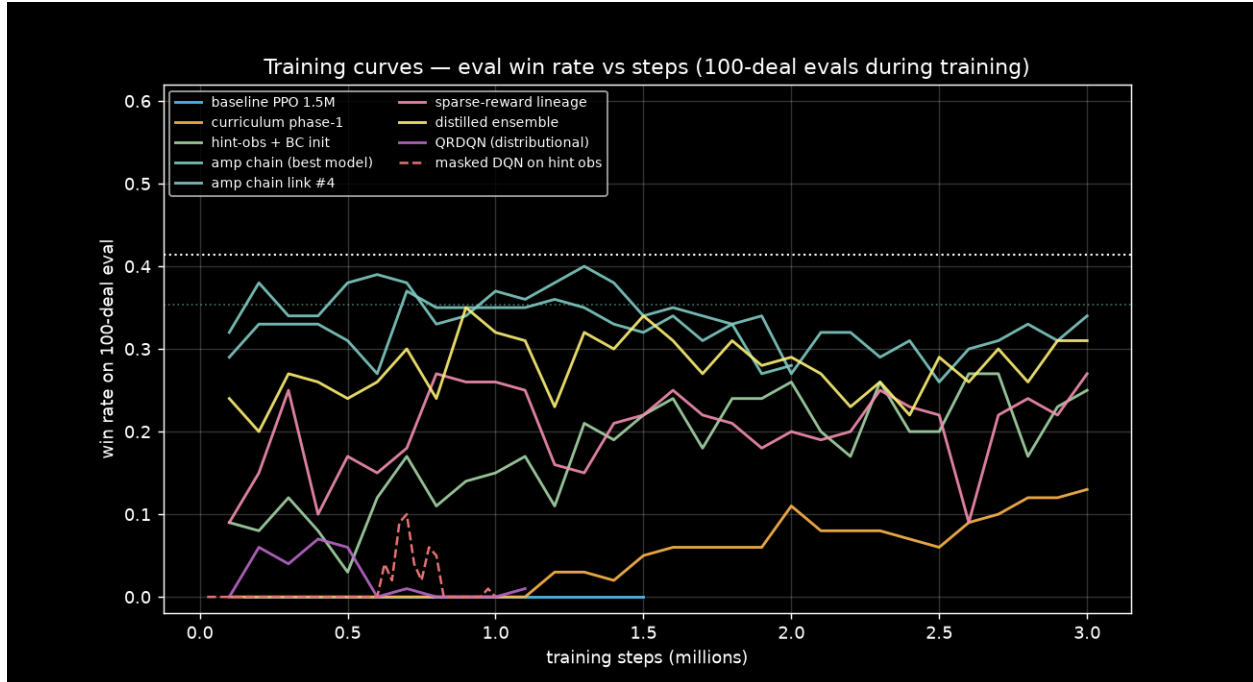


Figure 2: 100-deal eval win rate during training. The hint+BC lineage and the amplification chain dominate; curriculum stays flat; DQN/QRDQN collapse or never lift off; the dotted line is the heuristic’s full-1000 result.

Ceiling. A single model saturates at $\approx 35\%$ at this compute. The 37% of deals the portfolio still loses are a mix of provably unsolvable draws and deals needing deeper search than single-rollout policies provide.

6 Playable artifact

The repository ships a complete Klondike web game (FastAPI + vanilla JS): drag-and-drop with legal-target highlighting, undo, hints, sounds, auto-finish, and a *watch mode* in which any committed agent plays a chosen deal live. The backend uses the same engine and action masking as the evaluation harness, so watched games are exactly evaluated games. Figures, all eval JSONs, recorded winning episodes (GIFs/MP4), and a one-command runner (`./run_web.sh`) are included.

7 Related work

Klondike has been tackled with heuristic search and planning solvers; learned approaches are rarer, and to our knowledge none report a best-of-N lineage portfolio of RL agents on a fixed benchmark under laptop-scale compute. Our masked-value-method failures echo known difficulties of bootstrapping on long, sparse, partially observable episodes. The imitation+RL recipe is standard in spirit (e.g., expert demonstrations guiding exploration); our contribution is quantifying a concrete low-cost variant (Tanh-matched clone + visible expert action + champion-reclone chains) and showing where it saturates.

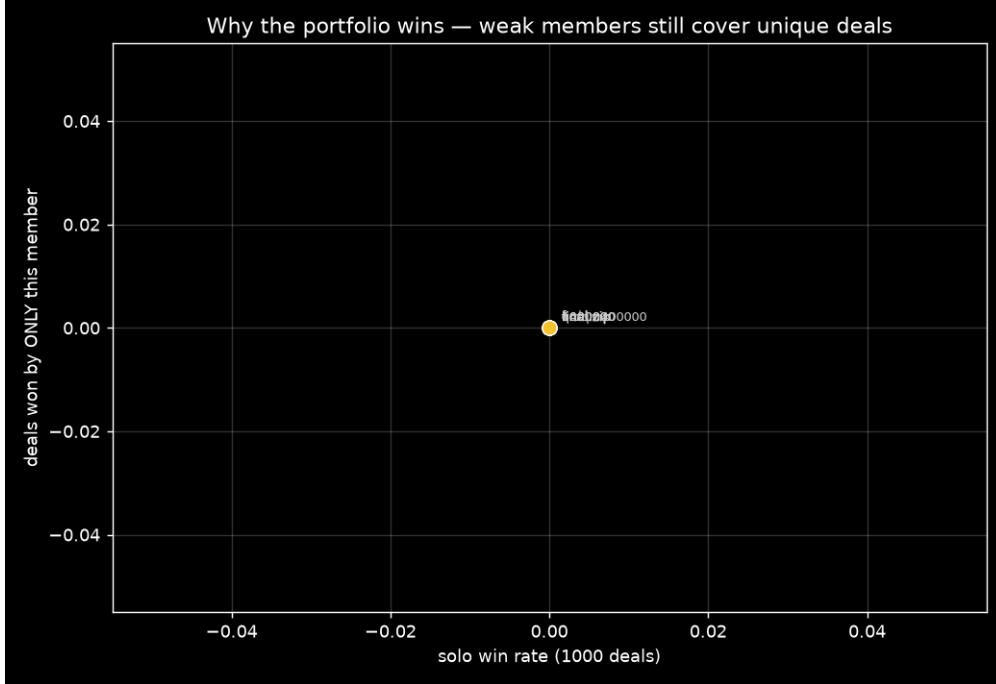


Figure 3: Each point is a portfolio member: solo win rate vs. deals won by *only* that member. Weak members (e.g. QRDQN at 6.7%) still cover deals no strong member wins — the mechanism behind the portfolio’s gains.

8 Limitations and future work

Single-rollout agents only; no MCTS or learned value-guided search at test time. The QRDQN bootstrap is unmasked (documented). Perfect-info models (+6 points) suggest a belief-state/history model could add more. The portfolio’s declared best-of-N is honest but offline — deploying it means running N rollouts per deal. Reproducing on draw-3 is open (heuristic itself drops to 11.8% there).

9 Conclusion

At laptop scale, masked PPO alone is weak on Klondike, and value-based methods fail outright. Imitation anchoring — an observation that keeps the expert visible, a shape-matched warm start, and re-cloning the strongest learned policy — lifts a single model to 35.3%. The best artifact, however, is a declared portfolio of 26 diverse lineages at 62.5%, comfortably above the 41.4% heuristic, with every number reproducible from the repository.

Reproducibility

All code, benchmark seeds, checkpoints for the canonical 9-member portfolio (57.9% measured), eval JSONs, training metrics, figures, and the paper source are in the public repository. `eval.py` regenerates every number; `reproduce.sh` replays the full pipeline; 58 unit tests pin rules, legality masks, and determinism. Exact committed portfolio spec is printed in `checkpoints/README.md`.